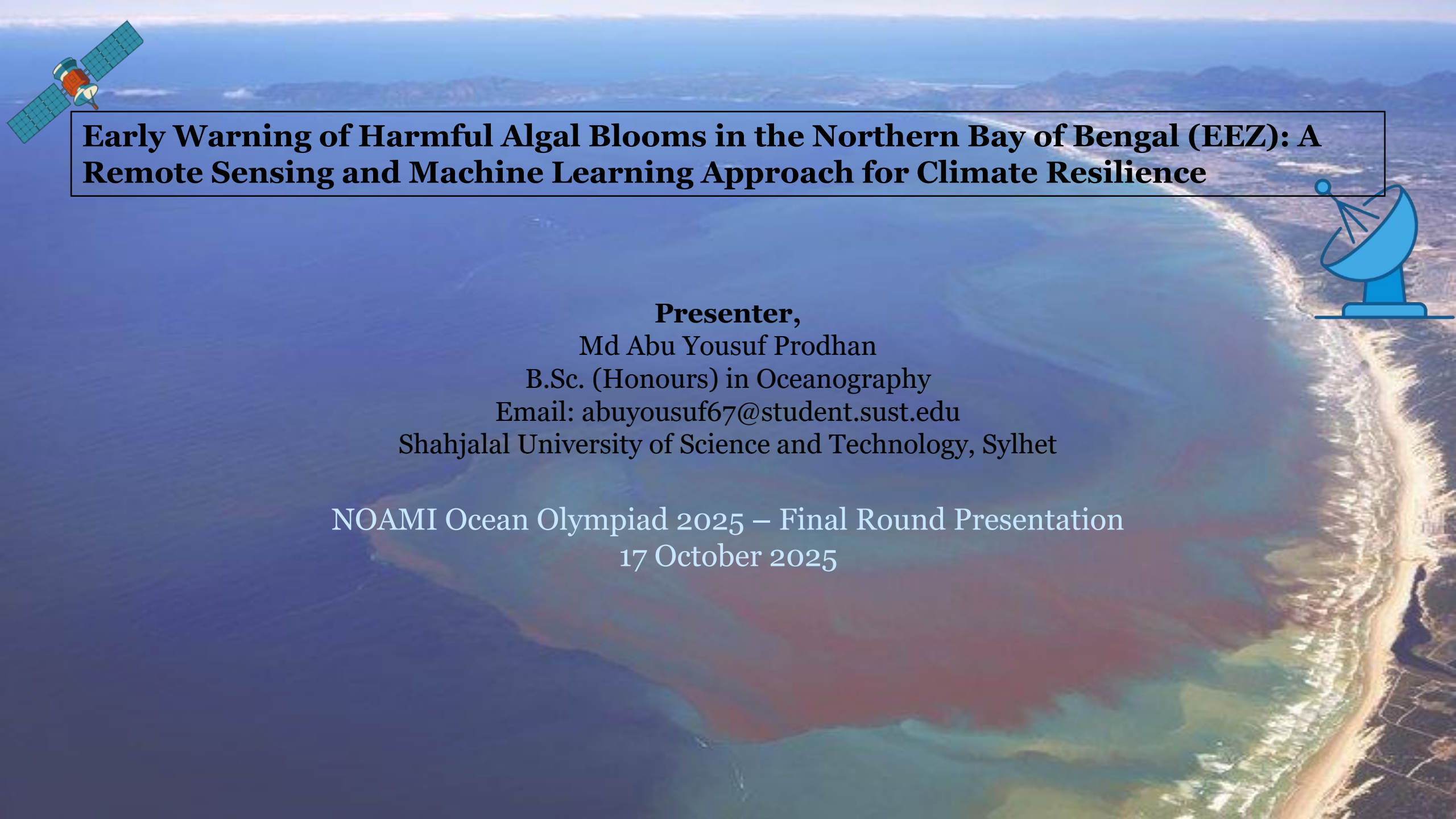




Early Warning of Harmful Algal Blooms in the Northern Bay of Bengal (EEZ): A Remote Sensing and Machine Learning Approach for Climate Resilience

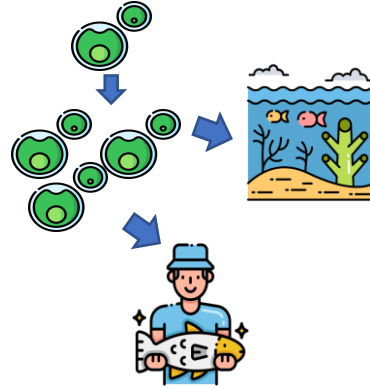


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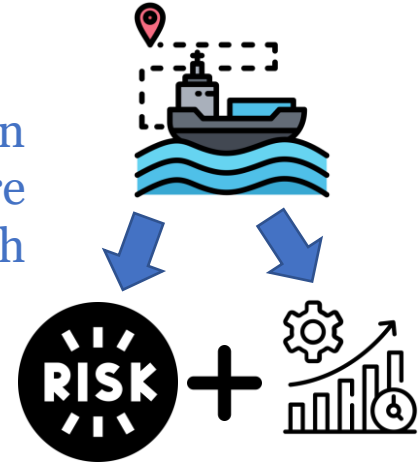
NOAMI Ocean Olympiad 2025 – Final Round Presentation
17 October 2025

Introduction

- ❑ Harmful Algal Blooms (HABs) are rapid growths of toxin-producing algae that threaten marine life and human health.



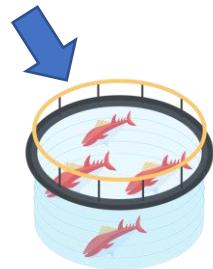
- ❑ HAB monitoring depends heavily on field-based observations, which are often not possible because of high expense and high risk areas.



- ❑ HABs directly impact-

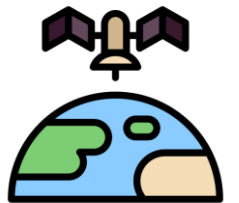


fisheries



aquaculture

- ❑ However, modern remote sensing technologies provide promising alternatives to overcome these limitations.



Introduction (cont..)

- ❑ Around 50 million coastal people in Bangladesh depend on marine resources.



- ❑ Over 35 million people directly rely on coastal fisheries for their livelihood.

- ❑ Artisanal fisheries supply 82% of marine catch and 60% of national protein.

- ❑ Around the world, HAB disasters have resulted in enormous financial losses.

➤ USA faced losses of USD \$5.6B in 2018 & 1.5B in 2017

➤ China faced losses of ¥ 1B during the years 2001, 2003 and 2015.



- ❑ Although HAB occurrences in the Bay of Bengal have increased (Khan et al., 2022)



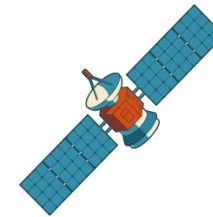
- ❑ Research and monitoring efforts remain limited



- ❑ No early warning or predictive HAB system exists in Bangladesh.



- ❑ this study addresses this gap by using-



satellite data



machine learning

Study Area

- ❑ Bangladesh has a **734 km long coastline**, supporting **~50 million people** — nearly **one-third of the population**.
- ❑ The study focuses on the **Exclusive Economic Zone (EEZ)** of Bangladesh.
- ❑ Geographic range: **20.70°N – 23°N** and **89°E – 92.46°E**.
- ❑ Total area: **~118,813 km²**.
- ❑ This zone is **ecologically rich** and vital for **fisheries and aquaculture**.
- ❑ Selected due to **frequent HAB events** and **absence of a local forecasting system**.

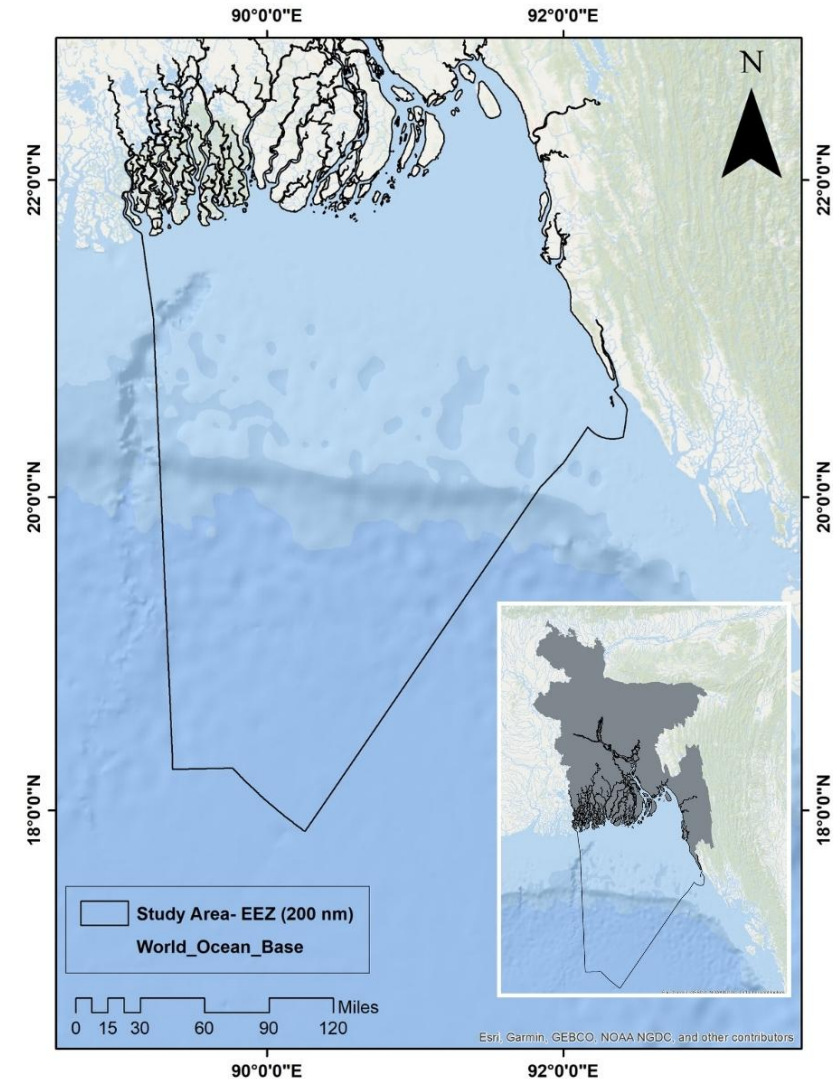


Figure 1: Map of the Study Area (EEZ) in the Northern Bay of Bengal

Aims & Objectives



To develop a **machine learning–based model** to predict **Harmful Algal Blooms (HABs)** in the **northern Bay of Bengal** using **satellite remote sensing data**.

Objectives



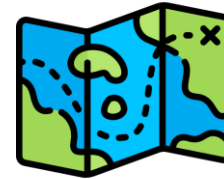
**Retrieve &
compile
MODIS
datasets**



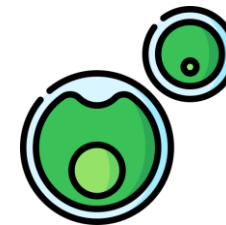
**Identify
key
variables**



**Train &
validate ML
models**



**Produce
HAB
risk maps**



**Estimate
algae cell
density**



**Design early
warning
systems**

Data & Variables

Data Acquisition



MODIS-Aqua **Level-3** ocean color satellite data (NASA Ocean Color Web)



Time period: 2010–2024



Resolution: Daily, 4 km



Coverage: Northern Bay of Bengal EEZ (~200 nm)

Key Parameters

Chlorophyll-a concentration (Chl-a_OCx)

Sea Surface Temperature (SST_4 or SST_11)

Fluorescence Line Height (FLH)

Diffuse Attenuation Coefficient (Kd_490_)

Euphotic Depth (Zeu)

Particulate Backscattering Coefficient (bbp_547)

Data Processing & Labelling

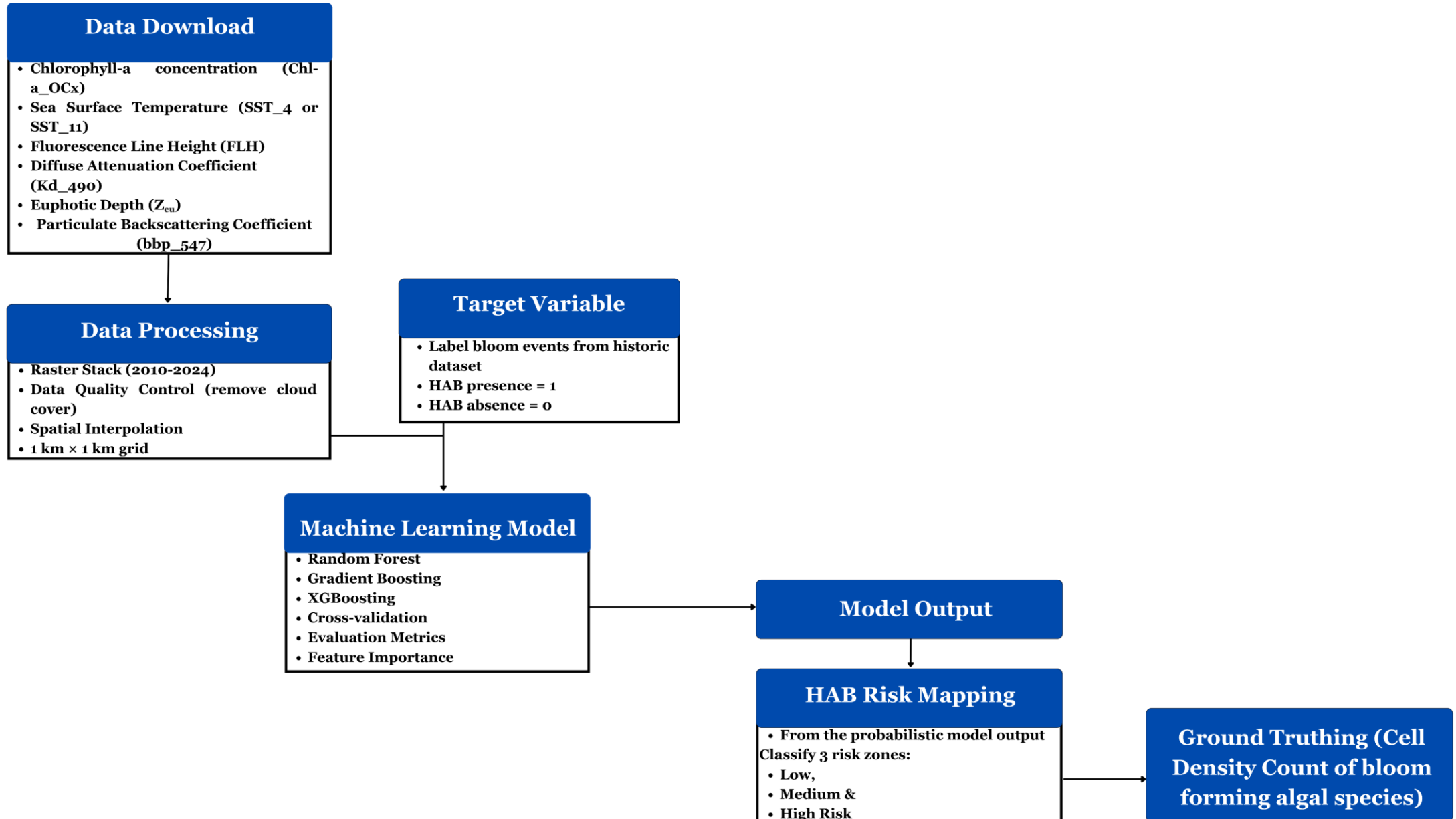
Build a **time-series raster stack** from daily L3 data

Filter datasets with cloud cover or missing data

Label HAB events:  1 = HAB occurrence (matched with historical records)

 0 = No bloom event

Methodology



Expected Outcomes

 **Machine learning-based model** to forecast HAB risk 5–7 days in advance.

 **Monthly & seasonal risk maps** (low, medium, high) to support **early warning & decision-making**.

 Identification of **key environmental drivers** (e.g., Chl-a, SST, FLH, Kd490).

 **Scalable framework** for other estuaries and coastal zones.

 **SMS-based early warning system** for aquaculture & fisheries communities.

 **Policy-relevant tool** for agencies (DoF, NGOs, WorldFish) to support **national HAB monitoring and management**.

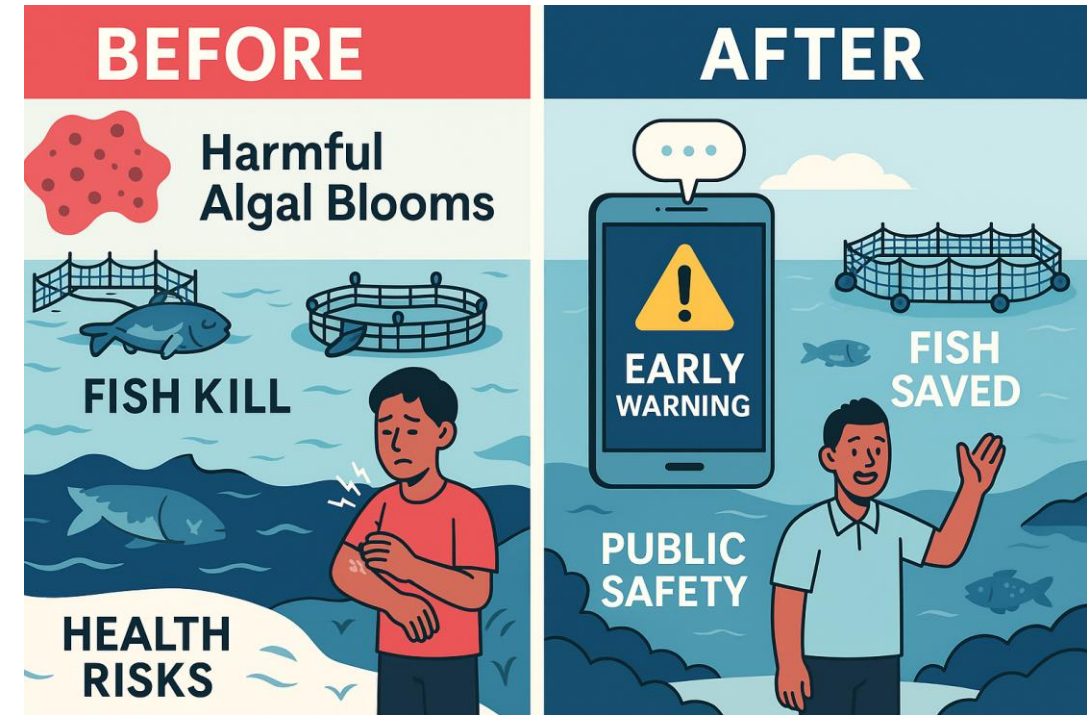
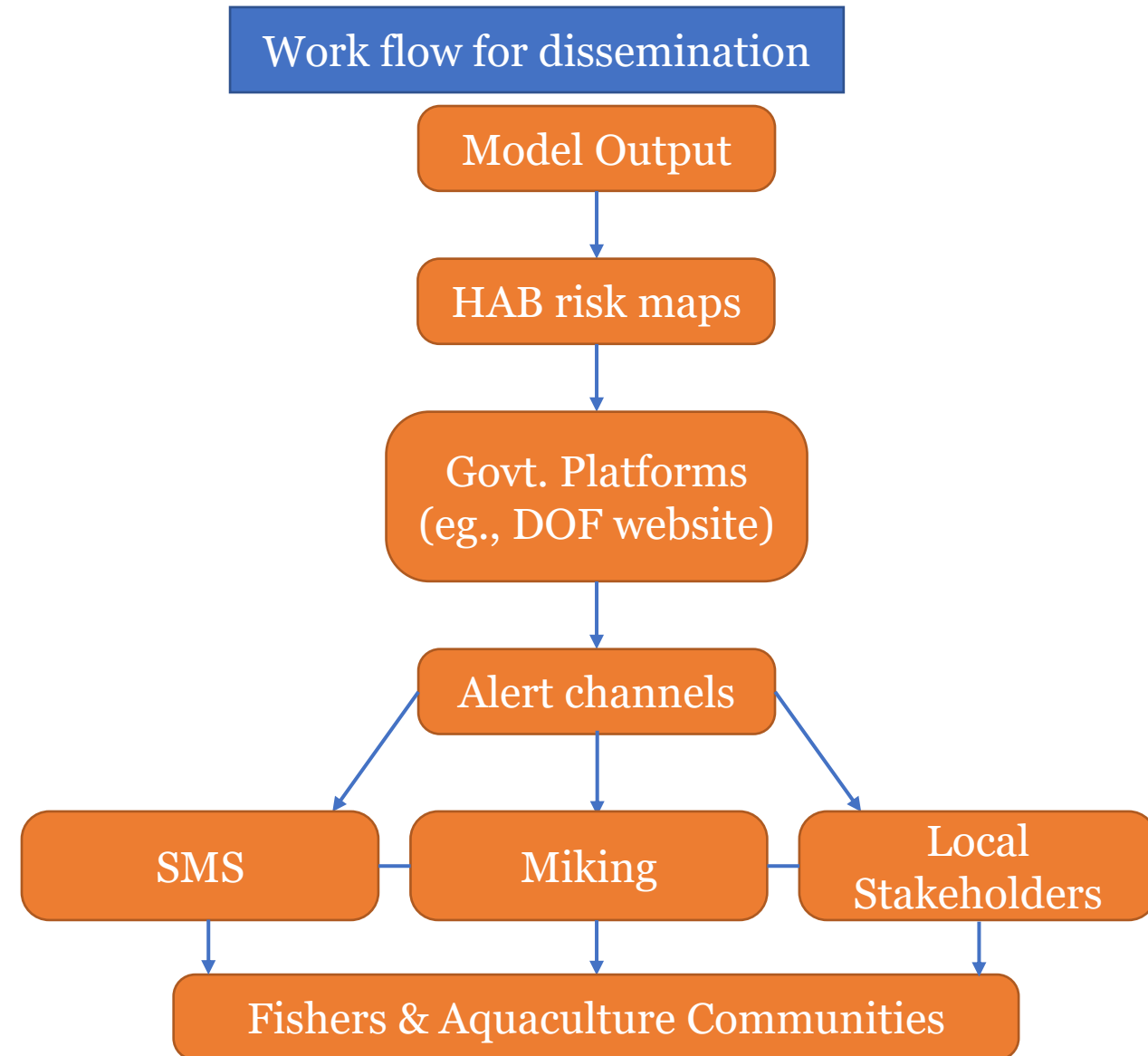


Figure 2: Before-and-after scenario showing how the HAB early warning system improves coastal preparedness and pollution response.

Community Impact & Dissemination

- ❑ Early warnings enable fishers & aquaculture stakeholders:
 - ✔ to avoid bloom-affected areas
 - ✔ plan fishing activities better
 - ✔ minimize economic losses
- ❑ Bloom-prone zones identified in advance are **communicated quickly** to vulnerable communities.
- ❑ Warnings delivered through **SMS, village miking**, and **local agencies** for wide coverage.
- ❑ Integrated into **Department of Fisheries website**, ensuring **national & local dissemination**.



Budget Breakdown

Serial	Item	Time	Monthly Cost (BDT)	Final Cost (BDT)
1	Software licenses / GIS tools	One-time	-	20,000
2	Historical data access & Literature buy	One-time	-	15,000
3	Printing, backup storage, internet	6 months	3,000	18,000
4	Ground truthing / validation <i>(optional)</i>	One-time	-	45,000
5	Contingency [Unforeseen costs]		-	15,000
Total	One Lac Thirteen Thousand BDT only			113,000

Conclusion

- ❑ **First ML-based HAB forecasting model** for the northern Bay of Bengal (Bangladesh EEZ).
- ❑ Integrates **remote sensing & machine learning** for low-cost, scalable, and accurate early warnings.
- ❑ Provides a **decision-support tool** for agencies (DoF, BORI) toward a **national HAB monitoring system**.
- ❑ Enhances **climate resilience and livelihood protection** for coastal fishers and aquaculture communities.
- ❑ Empowers local communities through **SMS alerts, citizen science programs, and awareness training**.
- ❑ **Promotes women's participation** in community-based HAB awareness and decision-making, advancing gender equity.
- ❑ Contributes to **Bangladesh's Delta Plan 2100, NAP, and SDGs 2, 5, 13, and 14**.

Alignment with SDGs



An aerial photograph of a coastline. The ocean is a deep blue, while the water near the shore is a mix of green and brown, indicating a large oil spill. The shoreline is visible on the right, with a sandy beach and some buildings. In the background, there are mountains under a clear blue sky.

Thank you